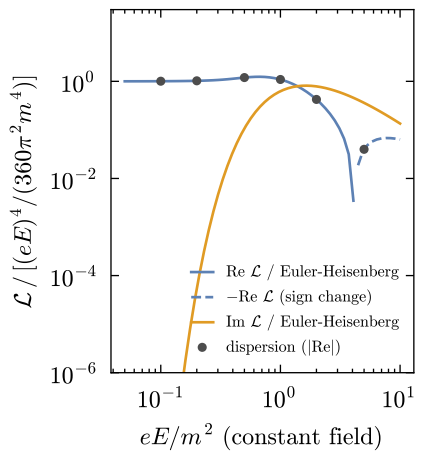
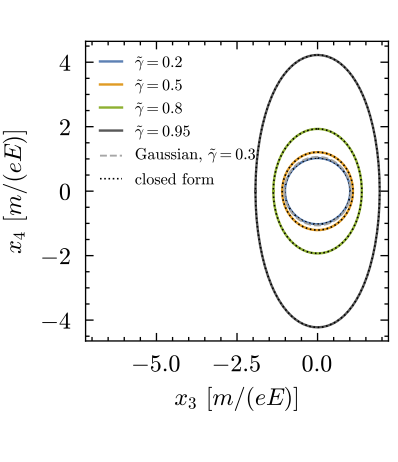
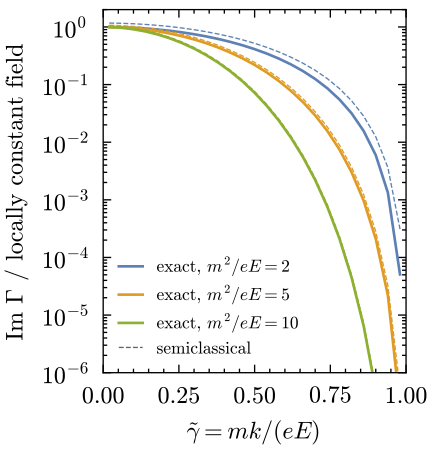

$\mathcal{L}\,/\,[(eE)^4/(360\pi^2 m^4)]$
10^0
10^{-2}
10^{-4}
10^{-6}
Re $\mathcal{L}$ / Euler-Heisenberg
$-$Re $\mathcal{L}$ (sign change)
Im $\mathcal{L}$ / Euler-Heisenberg
dispersion (|Re|)
10^{-1}
10^0
10^1
eE/m^2 (constant field)


$\tilde{\gamma} = 0.2$
$\tilde{\gamma} = 0.5$
$\tilde{\gamma} = 0.8$
$\tilde{\gamma} = 0.95$
Gaussian, $\tilde{\gamma} = 0.3$
closed form
$x_4\ [m/(eE)]$
4
2
0
-2
-4
-5.0
-2.5
0.0
$x_3\ [m/(eE)]$


Im Γ / locally constant field
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
exact, $m^2/eE = 2$
exact, $m^2/eE = 5$
exact, $m^2/eE = 10$
semiclassical
0.00
0.25
0.50
0.75
1.00
$\tilde{\gamma} = mk/(eE)$